```
C:\Windows\system32>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a              Displays current ARP entries by interrogating the current
                  protocol data.  If inet_addr is specified, the IP and Physical
                  addresses for only the specified computer are displayed.  If
                  more than one network interface uses ARP, entries for each ARP
                  table are displayed.
  -g              Same as -a.
  -v              Displays current ARP entries in verbose mode.  All invalid
                  entries and entries on the loop-back interface will be shown.
  inet_addr       Specifies an internet address.
  -N if_addr      Displays the ARP entries for the network interface specified
                  by if_addr.
  -d              Deletes the host specified by inet_addr. inet_addr may be
                  wildcarded with * to delete all hosts.
  -s              Adds the host and associates the Internet address inet_addr
                  with the Physical address eth_addr.  The Physical address is
                  given as 6 hexadecimal bytes separated by hyphens. The entry
                  is permanent.
  eth_addr        Specifies a physical address.
  if_addr         If present, this specifies the Internet address of the
                  interface whose address translation table should be modified
                  If not present, the first applicable interface will be used.

Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.

C:\Windows\system32>
```

Date: 25/7/26 Exp-1

Study of Various Network Commands used in Linux and Windows:

(1) arp-a:

ARP is short form of address resolution protocol. It will show the IP address of your computer along with the IP address and MAC address of your router.

(2) hostname:

It simply displays the name of your computer.

(3) IPconfig/all:

This command displays detailed configuration about your TCP/IP connection including router, gateway, DNS and type of Ethernet adapter in your system.

(4) nbstat-a:

This command helps solve problems with NetBIOS Name resolution.

(5) netstat:

It displays a variety of statistics about a computers active TCP/IP connection

eg netstat -r

```
C:\Windows\system32>ipconfig

Windows IP Configuration


Ethernet adapter Npcap Loopback Adapter:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::b923:c6f7:7122:8700%6
   Autoconfiguration IPv4 Address. . : 169.254.158.128
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::390c:dc1c:3ca7:94bc%3
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 172.16.10.86
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1

C:\Windows\system32>_
```

```
C:\Windows\system32>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.


C:\Windows\system32>
```

⑥ nslookup:

It is tool used to Perform DNS lookups in Linux. It is used to display DNS details, such as IP address of particular Computer, the MX records for a domain or the NS servers of domain

Eg nslookup www.google.com

⑦ PathPing:

It is unique to Window's and is basically a combination of Ping and Tracert Commands

⑧ Ping:

It is best way to test connectivity between two nodes. It use ICMP (Internet Control Message Protocol) to communicate to other devices.

① # Ping hostname (Ping localhost)
PingIPaddress (Ping 4.2.2.2)
Ping fully qualified domainname (
Ping www.facebook.com

⑨ Route:

Route command is used to show / manipulate the IP routing table. It is primarily

```
C:\Windows\system32>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49675        DESKTOP-G0LVE4J:49676  ESTABLISHED
  TCP    127.0.0.1:49676        DESKTOP-G0LVE4J:49675  ESTABLISHED
  TCP    127.0.0.1:49677        DESKTOP-G0LVE4J:49678  ESTABLISHED
  TCP    127.0.0.1:49678        DESKTOP-G0LVE4J:49677  ESTABLISHED
  TCP    172.16.10.86:13133     4.213.25.242:https     ESTABLISHED
  TCP    172.16.10.86:13199     lcbomp-in-f188:5228    ESTABLISHED
  TCP    172.16.10.86:13258     a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.10.86:13259     150.171.110.82:https   CLOSE_WAIT
  TCP    172.16.10.86:13260     131.253.33.254:https   CLOSE_WAIT
  TCP    172.16.10.86:13261     135.233.41.222:https   CLOSE_WAIT
  TCP    172.16.10.86:13262     204.79.197.222:https   CLOSE_WAIT
  TCP    172.16.10.86:13270     72.154.187.226:https   CLOSE_WAIT
  TCP    172.16.10.86:13272     20.100.121.19:https    CLOSE_WAIT
  TCP    172.16.10.86:13273     a-0003:http            CLOSE_WAIT
  TCP    172.16.10.86:13278     13.107.42.254:https    CLOSE_WAIT
  TCP    172.16.10.86:13279     52.108.9.254:https     CLOSE_WAIT
  TCP    172.16.10.86:13281     172.202.65.254:https   CLOSE_WAIT
  TCP    172.16.10.86:13293     a23-211-142-112:https  TIME_WAIT
  TCP    172.16.10.86:13294     a23-211-142-112:https  TIME_WAIT
  TCP    172.16.10.86:13295     a23-211-142-112:https  TIME_WAIT
  TCP    172.16.10.86:13297     bkk02s04-in-f14:https  ESTABLISHED
  TCP    172.16.10.86:13298     4.213.25.242:https     ESTABLISHED
  TCP    [fe80::390c:dc1c:3ca7:94bc%3]:1521  DESKTOP-G0LVE4J:49679  ESTABLISHED
  TCP    [fe80::390c:dc1c:3ca7:94bc%3]:49679  DESKTOP-G0LVE4J:1521   ESTABLISHED

C:\Windows\system32>_
```

C:\Windows\system32>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t   Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a   Resolve addresses to hostnames.
    -n count   Number of echo requests to send.
    -l size   Send buffer size.
    -f   Set Don't Fragment flag in packet (IPv4-only).
    -i TTL   Time To Live.
    -v TOS   Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the Type Of Service field in the IP
                   Header).
    -r count   Record route for count hops (IPv4-only).
    -s count   Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout   Timeout in milliseconds to wait for each reply.
    -R   Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr   Source address to use.
    -c compartment Routing compartment identifier.
    -p   Ping a Hyper-V Network Virtualization provider address.
    -4   Force using IPv4.
    -6   Force using IPv6.

```
C:\Windows\system32>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f            Clears the routing tables of all gateway entries.  If this is
                used in conjunction with one of the commands, the tables are
                cleared prior to running the command.

  -p            When used with the ADD command, makes a route persistent across
                boots of the system. By default, routes are not preserved
                when the system is restarted. Ignored for all other commands,
                which always affect the appropriate persistent routes.

  -4            Force using IPv4.

  -6            Force using IPv6.

  command       One of these:
                  PRINT     Prints  a route
                  ADD       Adds    a route
                  DELETE    Deletes a route
                  CHANGE    Modifies an existing route
  destination   Specifies the host.
  MASK          Specifies that the next parameter is the 'netmask' value.
  netmask       Specifies a subnet mask value for this route entry.
                If not specified, it defaults to 255.255.255.255.
  gateway       Specifies gateway.
  interface     the interface number for the specified route.
  METRIC        specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.
```

Result:

The networking commands were successfully executed and studied their outputs were observed and used to understand network config connectivity testing DNS resolution and trouble shooting